

REopt

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Results for Your Site



These results from REopt summarize the most cost-effective combination of PV, wind, battery storage, CHP, prime generator and/or emergency generator designed to sustain a critical load at your site. You can edit your inputs to see how changes to your energy strategies affect the results.



Your optimized solar installation size

1,810 kW

PV size

(0 kW new + 1,810 kW existing)

Measured in kilowatts (kW) of direct current (DC), this optimized size minimizes the life cycle cost of energy at your site.



Your optimized battery power and capacity

531 kW

battery power

3,968 kWh

battery capacity

The battery power (kW-AC) and energy capacity (kWh) are optimized independently for economic performance.

The optimized size may not be commercially available. The user is responsible for finding a commercial product that is closest in size to the optimized size.



Your potential life cycle savings (25 years)

This is the net present value of the savings (or costs if negative) realized by the project based on the difference between the total life cycle costs of doing business as usual compared to the optimal case.

\$1,222,221

[View citation](#)



Your Potential Resilience

This system sustains the 25% critical load during the specified outage period from July 1 starting at 0 am for 72 hours



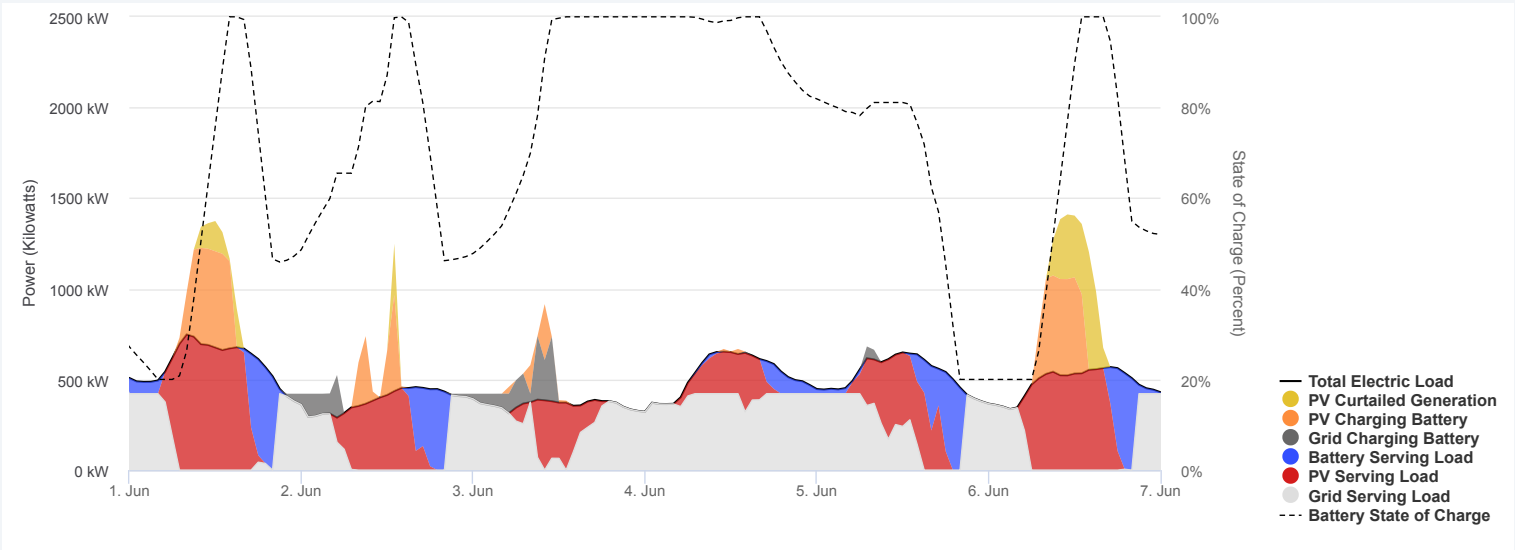
System survives specified outages



System Performance Year One

System Performance Year One

This interactive graph shows the dispatch strategy optimized by REopt for typical operation of the optimized system for every hour of the year. Graphs showing the optimized dispatch strategy during specified outages can be found in the Resilience vs. Financial tab below. To zoom in on a date range, click and drag right in the chart area or use the "Zoom In a Week" button. To zoom out, click and drag left or use the "Zoom Out a Week" button.



Annual Electricity Production Breakdown

This data shows the annual production of electricity for each technology and the breakdown of where the electricity goes. For PV and Wind, this is the average annual production which includes the year-over-year degradation included in the model.

Grid

Average Annual Dispatch Results	
Grid Serving Load (kWh)	1,744,298
Grid Charging Battery (kWh)	262,194
Grid Total Electricity Consumed (kWh)	2,006,492

PV

Average Annual Dispatch Results

PV Serving Load (kWh)	1,441,306
PV Charging Battery (kWh)	597,332
PV Exported to Grid (kWh)	0
PV Curtailment (kWh)	146,184
PV Total Electricity Produced (kWh)	2,184,822

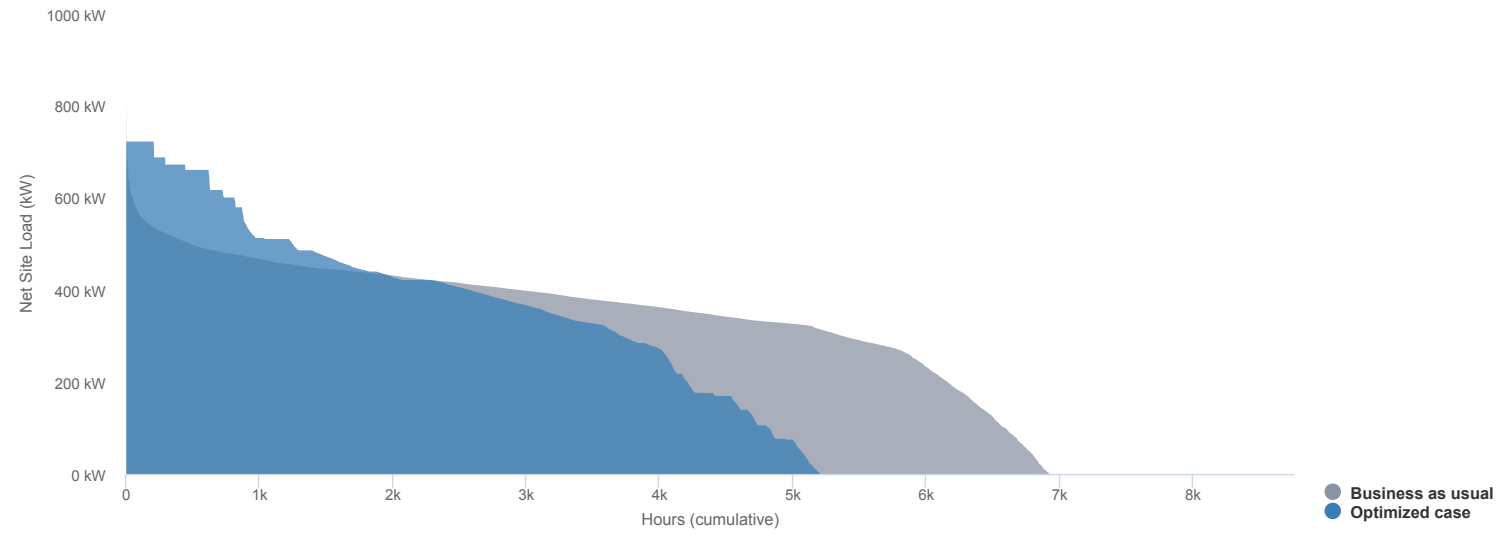
Battery

Average Annual Dispatch Results

Battery Serving Load (kWh)	773,464
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Net Load Duration

This interactive graph shows the reduction in peak load that occurs when the REopt optimized technologies are implemented. To zoom in on a date range, click and drag right in the chart area. To zoom out, click and drag left or use the "Reset zoom" button.



[Download Load Duration Spreadsheet](#)

[Download All Dispatch Data](#)

[Download All Annual Energy Production Breakdown Data](#)

\$Resilience vs. Financial

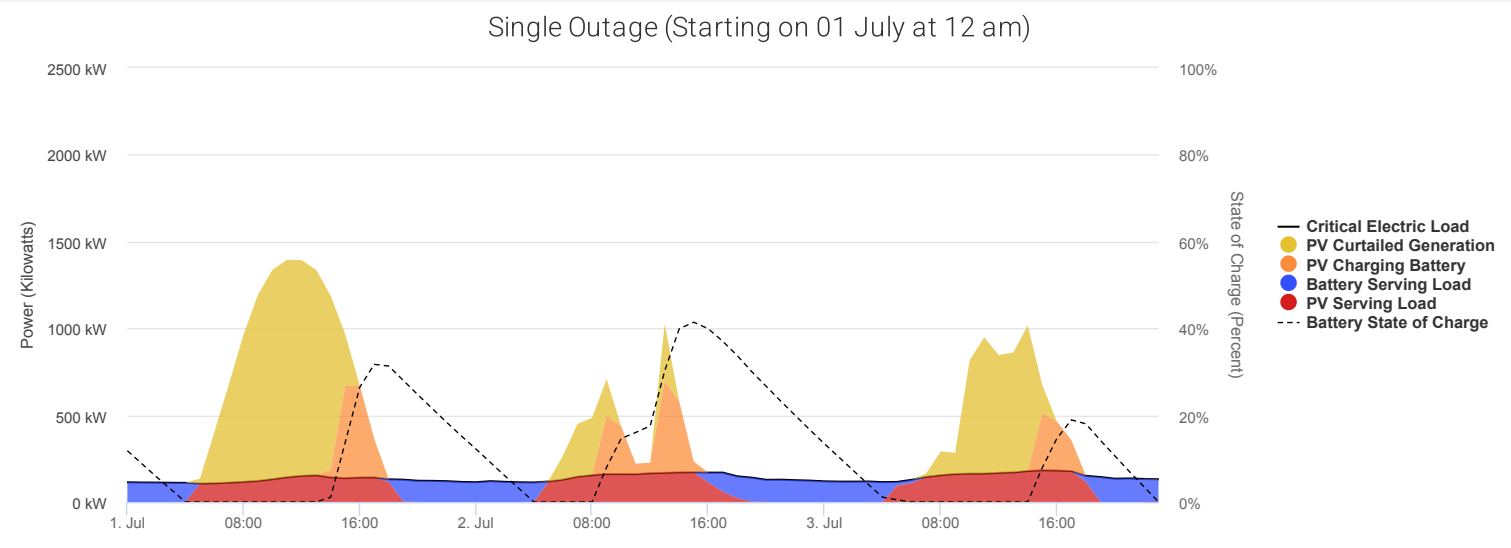
Resilience vs. Financial Comparison

This system was designed to sustain the critical load during the outage period specified at lowest cost. The Resilience vs. Financial Comparison table shows the architecture and cost differences between the Resilience system and the Financial system that does not consider resilience requirements. The DER technologies selected for the Resilience system and their sizes assume they are 100% available and reliable. To evaluate the probability of serving critical load considering variable reliability and for all possible outages in a year, go to the Energy Resilience Performance (ERP) section below.

	Business As Usual	Resilience	Financial
System	1810 kW PV	1,810 kW PV 531 kW Battery 3,968 kWh Battery	1,810 kW PV 531 kW Battery 3,968 kWh Battery
NPV	\$0	\$1,222,221	\$1,222,221

Outage Dispatch Plots

This interactive graph shows the dispatch strategy optimized by REopt for operation of the optimized system for every hour of an outage: either a single-outage period, if only one outage is modeled, or the specified outage that has been chosen using the corresponding radio buttons for multiple-outage evaluations. To zoom in on a date range, click and drag right in the chart area. To zoom out, click and drag left.



Outage Electricity Production Breakdown

This data shows the summary of how the critical load is served for each outage, including a breakdown of where the electric production by technology goes.

PV

Average Annual Dispatch Results	
Outage 1: PV Serving Load (kWh)	5,862
Outage 1: PV Charging Battery (kWh)	3,852
Outage 1: PV Curtailment (kWh)	14,415

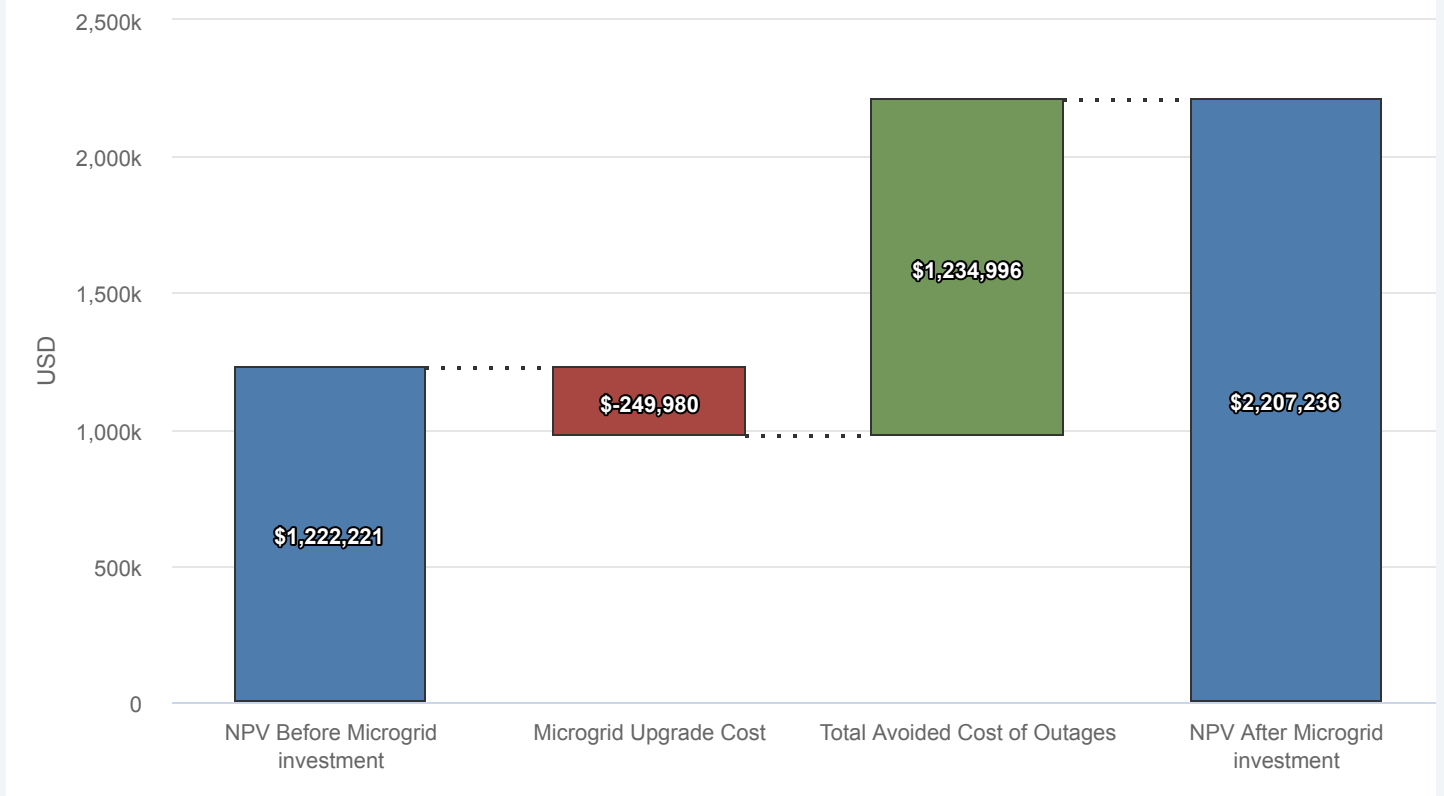
Outage 1: PV Total Electricity Produced (kWh)	24,130
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Battery

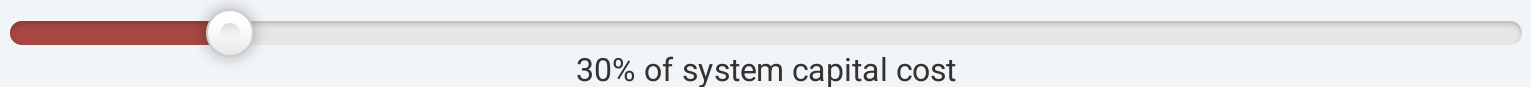
Average Annual Dispatch Results	
Outage 1: Battery Serving Load (kWh)	4,018

Effect of Resilience Costs and Benefits

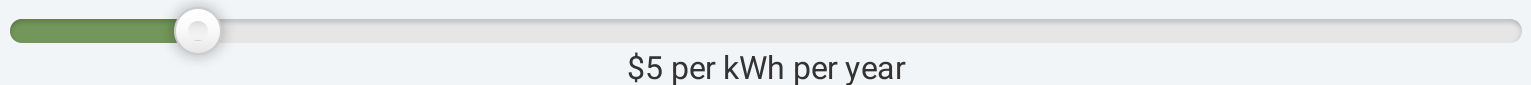
This interactive waterfall chart allows the user to consider the cumulative effect of extra costs and benefits of increased resilience on the project's net present value (NPV). Upgrading the optimized system to a microgrid allows a site to operate in both grid-connected and island-mode. This requires additional investment, which may include extra equipment such as controllers, distribution system infrastructure and communications upgrades. Economic benefit is observed when the value of avoiding the costs of an outage are considered. These microgrid upgrade costs and avoided outage costs are not factored into the optimization results. The sliders under the chart allow the user to change the Microgrid Upgrade Cost and the Annual Avoided Cost of Outage to analyze the impact on the NPV after Microgrid Costs and Benefits, while the NPV Before Microgrid Investment, which is determined by the optimization results, remains static.



Microgrid Upgrade Cost



Avoided Outage Cost Per kWh



Net capital cost to which the microgrid upgrade cost applies: \$833,268

Microgrid Upgrade Cost: $\$833,268 \times 30\% = \$249,980$

Average annual critical load served: 9,879 kWh

Analysis period years over which avoided outage cost applies: 25 years

Avoided Outage Costs: $\$5/\text{kWh}/\text{year} \times 9,879 \text{ kWh} \times 25 \text{ years} = \$1,234,995$

💡 Energy Resilience Performance

Energy Resilience Performance

The Energy Resilience Performance (ERP) of the solution is assessed based on the reliability of assets included for all possible outage periods in a year. The SOC for the start of an outage any time in the year is based on the optimization dispatch results. You can also modify the system architecture to see how

reliability performance might change. Common changes to consider are increasing the the nameplate capacity per unit.

Distributed Energy Resources	Total nameplate capacity (Optimized)	Number of units	Nameplate capacity per unit	Total nameplate capacity
Battery Power (kW)	531 kW	1	531 kW	531 kW
Battery Energy (kWh)	3,968 kWh	1	3968 kWh	3,968 kWh
PV (kW DC)	1,810 kW DC	1	1810 kW DC	1,810 kW DC

Max outage duration (hours)

72

Battery

Availability (%)

97%

PV

Availability (%)

98%

Results Comparison

These results show how doing business as usual compares to the optimal case.

	Business As Usual	Resilience	Financial
System Size			
PV Size	1,810 kW	1,810 kW	1,810 kW
Battery Power	0 kW	531 kW	531 kW
Battery Capacity	0 kWh	3,968 kWh	3,968 kWh
Energy Production and Fuel Use			
Average Annual PV Energy Production	2,184,822 kWh	2,184,822 kWh	2,184,822 kWh
Average Annual Energy Supplied from Grid	2,501,464 kWh	2,006,492 kWh	2,006,492 kWh
Renewable Energy Metrics			
Annual Renewable Electricity (% of electricity consumption)	37%	50%	50%
Climate Emissions			
Avoided CO ₂ e Emissions throughout Analysis Period	N/A	1,981 tonnes	1,981 tonnes
Health Emissions			
Avoided NO _x Emissions throughout Analysis Period	N/A	3.14 tonnes	3.14 tonnes
Avoided SO ₂ Emissions throughout Analysis Period	N/A	3.64 tonnes	3.64 tonnes

Avoided PM _{2.5} Emissions throughout Analysis Period	N/A	0.38 tonnes	0.38 tonnes
Year 1 Utility Electricity Cost — Before Tax			
Utility Export Benefit	\$0	\$0	\$0
Utility Energy Cost	\$109,647	\$78,790	\$78,790
Utility Demand Cost	\$153,206	\$69,548	\$69,548
Utility Fixed Cost	\$1,203	\$1,203	\$1,203
Utility Minimum Cost Adder	\$0	\$0	\$0
Total Year 1 Utility Cost - Before Tax	\$264,056	\$149,542	\$149,542
Life Cycle Cost Breakdown			
Technology Capital Costs + Replacements, After Incentives	\$0	\$833,268	\$833,268
O&M Costs	\$937,894	\$2,064,661	\$2,064,661
Total Utility Electricity Cost	\$7,337,910	\$4,155,653	\$4,155,653
Total Production-Based Incentive	\$0	\$0	\$0
Outage Fuel Costs	\$0	\$0	\$0
Cost of Climate Emissions throughout Analysis Period (If Included in Objective)	\$0	\$0	\$0
Cost of Health Emissions throughout Analysis Period (If Included in Objective)	\$0	\$0	\$0

Summary Financial Metrics

Total Upfront Capital Cost Before Incentives	\$0	\$1,739,599	\$1,739,599
Year 1 O&M Cost, Before Tax	\$36,200	\$79,690	79,690
Total Life Cycle Costs	\$8,275,804	\$7,053,582	\$7,053,582
Net Present Value	\$0	\$1,222,221	\$1,222,221
Payback Period	N/A	12.62 yrs	12.62 yrs
Internal Rate of Return	N/A	6.7%	6.7%
PV Levelized Cost of Energy	N/A	\$0.000/kWh	\$0.000/kWh

 Renewable Energy & Emissions Metrics

Renewable Energy & Emissions Metrics

These results show emissions outcomes for the business as usual and optimized cases. If marginal grid emissions rates are utilized (the default inputs), users should focus on avoided emissions, rather than emissions totals. Note for all emissions outputs, “t” (as in, "t CO₂e") represents metric tons (tonnes).

	Business As Usual	Resilience	Financial
Renewable Energy			
Annual Renewable Electricity (% of electricity consumption)	37%	50%	50%
Climate & Health Emissions Costs			
Cost of Climate Emissions throughout Analysis Period	\$863,686	\$683,062	\$683,062

Cost of Health Emissions throughout Analysis Period	\$2,549,943	\$2,028,050	\$2,028,050
Climate Emissions, CO ₂ e			
Average Annual Emissions (t CO ₂ e)	379	300	300
Average Annual Emissions from Grid Purchases (t CO ₂ e)	379	300	300
Average Annual Emissions from Onsite Fuel Burn (t CO ₂ e)	0	0	0
Total Emissions throughout Analysis Period (t CO ₂ e)	9,473	7,492	7,492
Emissions from Grid Purchases throughout Analysis Period (t CO ₂ e)	9,473	7,492	7,492
Emissions from Onsite Fuel Burn throughout Analysis Period (t CO ₂ e)	0	0	0
Percent Reduction in CO2 Emissions from BAU (%)	N/A	20.91%	20.91%
Breakeven Cost of CO ₂ e Emissions Reduction (\$/t CO ₂ e)	N/A	N/A	N/A
Health Emissions, NO _x			
Average Annual Emissions (t NO _x)	0.60	0.48	0.00
Average Annual Emissions from Grid Purchases (t NO _x)	0.60	0.48	0.00

Average Annual Emissions from Onsite Fuel Burn (t NO _x)	0.00	0.00	0.00
Total Emissions throughout Analysis Period (t NO _x)	15.07	11.93	12.00
Emissions from Grid Purchases throughout Analysis Period (t NO _x)	15.07	11.93	12.00
Emissions from Onsite Fuel Burn throughout Analysis Period (t NO _x)	0.00	0.00	0.00
Health Emissions, SO ₂			
Average Annual Emissions (t SO ₂)	0.75	0.61	1.00
Average Annual Emissions from Grid Purchases (t SO ₂)	0.75	0.61	1.00
Average Annual Emissions from Onsite Fuel Burn (t SO ₂)	0.00	0.00	0.00
Total Emissions throughout Analysis Period (t SO ₂)	18.84	15.20	15.00
Emissions from Grid Purchases throughout Analysis Period (t SO ₂)	18.84	15.20	15.00
Emissions from Onsite Fuel Burn throughout Analysis Period (t SO ₂)	0.00	0.00	0.00
Health Emissions, PM _{2.5}			
Average Annual Emissions (t PM _{2.5})	0.07	0.05	0.00

Average Annual Emissions from Grid Purchases (t PM _{2.5})	0.07	0.05	5.00
Average Annual Emissions from Onsite Fuel Burn (t PM _{2.5})	0.00	0.00	0.00
Total Emissions throughout Analysis Period (t PM _{2.5})	1.66	1.28	128.00
Emissions from Grid Purchases throughout Analysis Period (t PM _{2.5})	1.66	1.28	1.00
Emissions from Onsite Fuel Burn throughout Analysis Period (t PM _{2.5})	0.00	0.00	0.00

[Download Hourly Grid Emissions Factors](#)

 Inputs

Your Inputs

The results are based on the following user supplied inputs.

Energy Goals

Cost-Savings

\$

Resilience



Technologies Selected

PV



Battery



Site

Evaluation name	test7 + Battery Capacity Unlimited
Site Location	Madison, WI, United States (43.07313, -89.38644)
PV & wind space available	Roofspace
Roofspace available (sq ft)	1,000,000,000

Utilities	
URDB rate	Wisconsin Electric Power Co (Michigan) - General Secondary Service Time-of-Use (Cg3)

Load Profile	
Typical electric load profile type	uploaded
Uploaded typical electric load profile	load_profile_tem (1).csv

Financial	
Analysis period (years)	25
Host discount rate, nominal (%)	0%
Electricity cost escalation rate, nominal (%/year)	3%

Renewable Energy & Emissions Accounting	
Cambium location	MISO North
Cambium Levelization Years	25

PV	
Existing PV systems size (kW-DC)	1810.0
Type of load profile	gross

Maximum new PV size (kW-DC)	0.0
PV generation profile	pv_profile_tem_norm.csv
Size class	Large Commercial (101 - 2,000 kW)

Battery	
Minimum energy capacity (kWh)	0.0
Minimum power capacity (kW)	300.0

Resilience	
Number of Outages	1
Outage start date	Jul 1
Outage start time	0 am
Outage duration (hours)	72
Critical load profile type	Percent
Critical load factor	25.0%

 Defaults

Default Inputs

The results are based on the following default inputs.

Site	
Sector	Commercial/Industrial
Solver optimality tolerance (%)	0.1%
Solver name	HiGHS
Optimization timeout (seconds)	600

Utilities	
Compensation type	No compensation for exports

Load Profile	
Load adjustment (%)	100%
Annual electric energy consumption (kWh)	3,959,068

Financial	
Host effective tax rate (%)	26%
O&M cost escalation rate (%/year)	2.5%
Minimum capital cost, before incentives (\$)	None
Maximum capital cost, before incentives (\$)	None
Third Party Ownership	No
Third-party owner discount rate, nominal (%)	6.24%
Third-party owner effective tax rate (%)	26%

Renewable Energy & Emissions Accounting	
EPA's AVERT Region	Midwest
Geographic resolution	GEA Regions 2023
Metric	LRMER CO2e Combined
Grid scenario	Mid-case
Use emissions averaged over the analysis period?	Yes
Cambium start year	2025
Include distribution losses?	Enduse
Projected annual percent decrease in grid	4.590%

health emissions factors (%/year)	
Count exported renewable electricity towards renewable energy goals?	true
Count exported electricity towards emissions offsets?	true
CO ₂ cost (\$/t CO ₂)	51.0
On-site fuel burn NOx cost (\$/t NOx)	30,061.54
On-site fuel burn SO ₂ cost (\$/t SO ₂)	51,345.83
On-site fuel burn PM2.5 cost (\$/t PM2.5)	355,946.69
Grid emissions NOx cost (\$/t NOx)	24,334.71
Grid emissions SO ₂ cost (\$/t SO ₂)	49,903.97
Grid emissions PM2.5 cost (\$/t PM2.5)	207,653.69
CO ₂ cost escalation rate, nominal (%)	4.22%
NOx cost escalation rate, nominal (%)	3.47%
SO ₂ cost escalation rate, nominal (%)	4.00%
PM2.5 cost escalation rate, nominal (%)	3.74%

PV	
System capital cost (\$/kW-DC)	\$1,920
O&M cost (\$/kW-DC per year)	\$20
Minimum new PV size (kW-DC)	0
Module type	Standard
Array azimuth (deg)	180
System losses (%)	14%
PV Station Search Radius (mi)	Unlimited
Rooftop power density (kW-DC per square	0.01

	foot)
Federal percentage-based incentive (%)	30%
Federal maximum incentive (%)	Unlimited
Federal rebate (\$/kW-DC)	\$0
Federal maximum rebate (\$)	Unlimited
State percentage-based incentive (%)	0%
State maximum incentive (\$)	Unlimited
State rebate (\$/kW-DC)	\$0
State maximum rebate (\$)	Unlimited
Utility percentage-based incentive (%)	0%
Utility maximum incentive (\$)	Unlimited
Utility rebate (\$/kW-DC)	\$0
Utility maximum rebate (\$)	Unlimited
Production incentive (\$/kWh)	\$0
Incentive duration (years)	1
Maximum incentive (\$)	Unlimited
System size limit (kW-DC)	Unlimited
MACRS bonus depreciation	100%
MACRS schedule	5 years

Battery	
Energy capacity cost (\$/kWh)	\$253
Power capacity cost (\$/kW)	\$968
Constant cost (\$)	\$222,115
Annual O&M cost as a percent of upfront	2.5%

	cost (%)
Allow grid to charge battery	Yes
Energy capacity replacement cost (\$/kWh)	\$0
Energy capacity replacement year	10
Power capacity replacement cost (\$/kW)	\$0
Power capacity replacement year	10
Constant replacement cost (\$)	\$0
Constant replacement year	10
Maximum energy capacity (kWh)	Unlimited
Maximum power capacity (kW)	Unlimited
Minimum battery duration (hours)	0
Maximum battery duration (hours)	Unlimited
Rectifier efficiency (%)	96%
Round trip efficiency (%)	97.5%
Inverter efficiency (%)	96%
Minimum state of charge (%)	20%
Initial state of charge (%)	50%
Total percentage-based incentive (%)	30%
Total power capacity rebate (\$/kW)	\$0
MACRS bonus depreciation	100%
MACRS schedule	5 years

Resilience
Critical load factor

Caution

Investment decisions should not be made on REopt results alone. These results assume perfect prediction of solar irradiance, wind speed, and electrical and thermal loads. In practice, actual savings may be lower based on the ability to accurately predict solar irradiance, wind speed, and load, and the control strategies used in the system. And, when modeling a grid outage the results assume perfect foresight of the impending outage, allowing the battery system to charge in the hours leading up to the outage. If a natural gas-fueled CHP system is included, the resiliency results assume the natural gas supply is not disrupted during an electrical grid outage.

The results include both expected energy and demand savings. However, the hourly model does not capture intra-hour variability of the PV and wind resource. Because demand is typically determined based on the maximum 15-minute peak, the estimated savings from demand reduction may be exaggerated. The hourly simulation uses one year of load data and one year of solar and wind resource data. Actual demand charges and savings will vary from year to year as load and resource vary.

Asset dispatch decisions are determined by the model as part of the cost-minimization objective. In application, some aspects of these operational decisions may not work well with the existing infrastructure or may not follow best practices. For example, in results with CHP, boiler dispatch may result in short cycling or periodic boiler use that is not possible without hot-standby. The user should review the dispatch results with these limitations in mind.

PV system performance predictions calculated by PVWatts include many inherent assumptions and uncertainties and do not reflect variations between PV technologies nor site-specific characteristics except as represented by inputs. For example, PV modules with better performance are not differentiated within PVWatts from lesser-performing modules.

Next Steps

Next Steps

This model provides an **estimate** of the techno-economic feasibility of solar, wind, battery, and/or CHP but investment decisions should not be made based on these results alone. **Before moving ahead with project development, verify:**

- The utility rate tariff is correct.

- Note that a site may have the option or may be required to switch to a different utility rate tariff when installing a renewable energy system.
- Contact your utility for more information.
- Actual load data is used rather than a simulated load profile.
- The load adjustment is entered as intended. (To learn more about achieving energy efficiency savings, visit the [Better Buildings Solution Center](#)).
- PV, wind, battery, and CHP costs and incentives are accurate for your location.
 - There may be additional value streams not included in this analysis such as ancillary services or capacity payments.
- Financial inputs are accurate, especially discount rate and utility escalation rate.
- Other factors that can inform decision-making, but are not captured in this model, are considered.

These may include:

- roof integrity
- shading considerations
- obstacles to wind flow
- ease of permitting
- mission compatibility
- regulatory and zoning ordinances
- utility interconnection rules
- availability of funding.
- Multiple systems integrators are consulted and multiple proposals are received. These will help to refine system architecture and projected costs and benefits. REopt results can be used to inform these discussions.

Contact NLR (formerly NREL) at reopt@nrel.gov for more detailed modeling and project development assistance.